

Prophylaxis and Treatment of Kidney Transplant Rejection

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The clinical presentation of the immune response to transplanted tissue (referred to as *rejection*) became apparent in 1960 when, after successful proof-of-principle kidney transplants were performed in identical twins, kidney transplantation was attempted between immunologically dissimilar individuals.¹ Eleven patients underwent lymphoid irradiation to prevent rejection after kidney transplant from nonidentical donors. Although 10 of 11 died of overwhelming infection, illustrating the potential consequences of immunosuppression, the lone surviving patient from this series subsequently experienced two episodes of acute rejection, both of which were treated with corticosteroids, and recovered good graft function. Thus, began the development of immunosuppressive agents that could prevent and treat rejection while not inducing severe life-threatening side effects. As a result of the development of newer immunosuppression medications, the incidence of acute rejection in the first year after transplant has significantly decreased over time and is now approximately 10% to 15%² (Fig. 109.1). However, the management of chronic rejection has remained a challenge, with continued attempts to better define the nature of injury and methods to prevent or reverse this process. As attention shifts to limiting the toxicity of immunosuppression medications (e.g., withdrawal of corticosteroids or calcineurin inhibitors [CNIs]) and attempts to increase access to transplant (e.g., performing transplants across human leukocyte antigen [HLA] and blood type barriers), management of acute rejection continues to be an important clinical issue.

DEFINITION

Rejection (both acute and chronic) is defined by histologic findings after kidney transplant biopsy. An adequate biopsy specimen contains at least 10 glomeruli and 2 small arteries, stained for hematoxylin-eosin (HE), periodic acid-Schiff (PAS) or methenamine silver, and Masson trichrome, and a specimen with seven to nine glomeruli and one artery is considered of marginal adequacy. When performed for clinical indications (kidney dysfunction), two separate cores should be obtained because the findings of rejection are often patchy in distribution (Fig. 109.2).³

The Banff Working Classification of Renal Allograft Pathology forms the basis of the histologic definition of rejection and is updated biannually (Boxes 109.1 and 109.2). First developed in 1993 with a primary focus on T-cell-mediated acute inflammatory infiltrates to classify the degree of rejection, updates in the classification differentiated humoral (antibody-mediated) and T-cell responses and further distinguished a chronic humoral form of injury previously classified as chronic allograft nephropathy (see Chapter 112).⁴ This was based on the indirect identification of antibody-mediated injury by evidence of complement (C4d) deposition. C4d is a fragment of C4b that is generated after immunoglobulin G (IgG) and IgM binding to host antigens with activation of the classical complement pathway. C4b/C4d forms a

covalent bond with proteins on tissue such as capillary endothelial cells via a sulfhydryl group and remains bound to tissue after immunoglobulin and other complement products have been released.⁵ Staining for C4d, either by immunohistochemistry or immunofluorescence, indicates that an antibody-mediated process may be involved.

Although C4d remains a specific marker for antibody-mediated graft pathology, it has limited sensitivity.⁶ Studies of endothelial-associated transcript (ENDAT) expression^{7,8} and C4d-negative histologic findings from protocol biopsy studies in patients with microvascular inflammation⁹ helped identify the clinical importance of C4d-negative antibody-mediated rejection, which led to these updated Banff criteria.

Antibody-Mediated Rejection

Acute antibody-mediated (humoral) rejection occurs in 5% to 7% of all transplants and is present in 20% to 30% of episodes of acute rejection¹⁰ in patients with ABO- and HLA-compatible transplants, occurring typically within the first few weeks of transplantation or in association with a change or poor adherence to immunosuppression. Although patients who have preexisting donor-specific HLA alloantibodies (donor-specific antibodies) are at higher risk for acute humoral rejection, the identification of *de novo* donor-specific antibodies at the time of graft dysfunction is common. There is also increasing recognition of the pathologic role of non-HLA antibodies directed against autoantigens on the vascular endothelium. The diagnosis of acute humoral rejection requires (1) identification of circulating donor-specific antibodies, (2) evidence that those antibodies are interacting with the vascular endothelium, and (3) evidence that this interaction is causing tissue injury (Box 109.1 and Fig. 109.3).⁶ Patterns of injury associated with acute humoral rejection range from acute tubular cell injury to thrombotic microangiopathy, but neutrophils and/or macrophages are typically present in peritubular capillaries. The requirement for evidence of antibody/endothelial interaction has been expanded beyond C4d staining and now includes at least moderate microvascular inflammation or evidence of increased ENDAT expression if using a validated assay.

Chronic active antibody-mediated rejection has been identified as a leading cause of late allograft loss^{7,11} and is likely the result of an indolent alloimmune response that can result in transplant glomerulopathy and microcirculatory inflammation. Although transplant glomerulopathy is often associated with circulating donor-specific antibodies and C4d deposition, these diagnostic markers are not detected in 30% to 50% of cases.¹² One possibility is that transplant glomerulopathy is not solely due to a humoral response. Alternatively, the failure to detect donor-specific antibodies and C4d deposition could be due to non-complement-fixing antibodies and/or the waxing/waning nature of the humoral response.

The current Banff diagnostic criteria for chronic humoral rejection require (1) evidence of donor-specific antibodies, (2) evidence of current/recent antibody interaction with vascular endothelium, and (3)

One-Year Incidence of Acute Rejection Over Time Associated With Immunosuppressive Agents

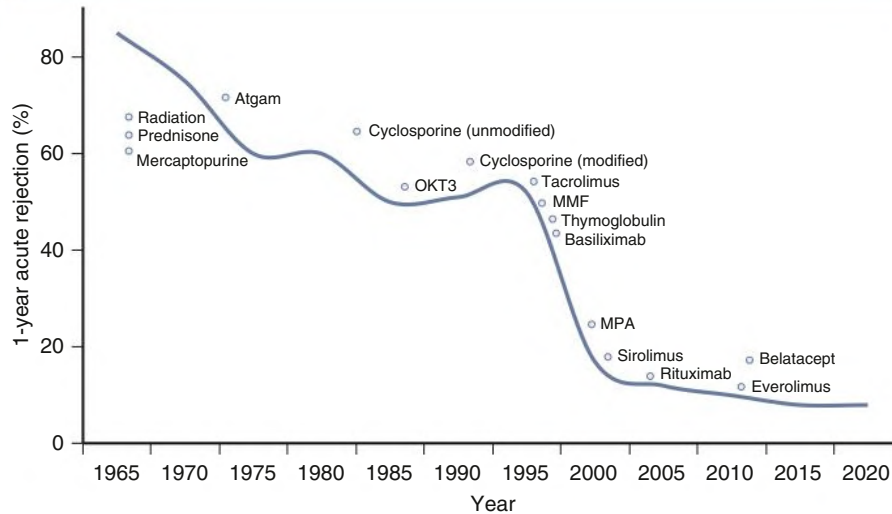


Fig. 109.1 One-year incidence of acute rejection over time as associated with the introduction of new immunosuppressive agents. *MMF*, Mycophenolate mofetil; *MPA*, mycophenolate sodium. (From Cooper JE. Evaluation and treatment of acute rejection in kidney allografts. *Clin J Am Soc Nephrol*. 2020;15[3]:430–438.)

Effect of Sample Site on the Diagnosis of Rejection

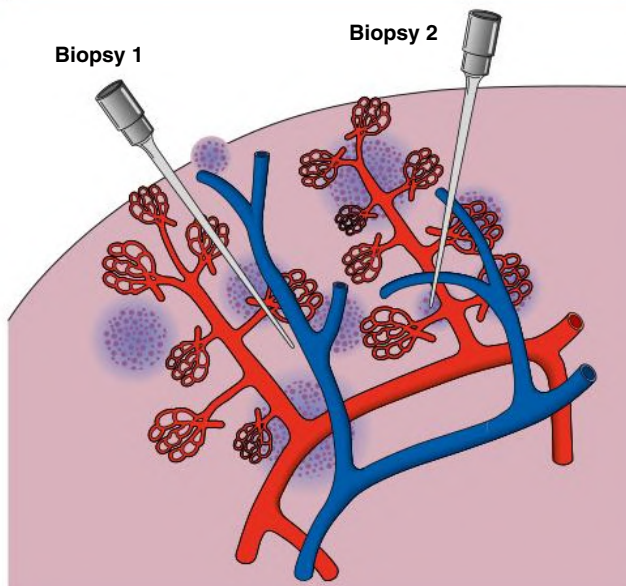


Fig. 109.2 Biopsy Technique. Acute rejection begins as patchy, focal infiltrates and becomes homogeneous only in advanced stages. The intensity of mononuclear infiltrate seen on the biopsy sample would differ between core 1 and core 2. Routinely taking two core biopsy samples can help decrease the sampling errors, which can affect the histologic interpretation of rejection.

evidence of chronic tissue injury such as transplant glomerulopathy, multilamination of the peritubular capillary basement membrane, and/or arterial intimal fibrosis (Fig. 109.4).

T-Cell–Mediated Rejection

The original pathologic description of rejection is now referred to as *T-cell–mediated rejection*. The classification of acute T-cell–mediated

rejection (acute cellular rejection [ACR]) is based on the degree and location of mononuclear cell inflammation. Because interstitial inflammation and tubulitis are frequently present immediately beneath the kidney capsule (subcapsular inflammation) in stable allografts, this finding is not considered when interpreting an allograft biopsy result for the presence of rejection. When severe, interstitial inflammation may extend into tubules via injury to the tubular basement membrane (tubulitis). The predominant phenotype of these infiltrates is a mixture of CD4⁺ and CD8⁺ T-cells; however, B-cells, eosinophils, and macrophages also may be present. Less commonly, endarteritis (endothelialitis) may be present in which T-cells and macrophages extend under the arterial endothelium, a phenomenon that may or may not be accompanied by interstitial inflammation or tubulitis. The finding of interstitial infiltrates and tubulitis in a kidney transplant biopsy is not specific to ACR, and other causes such as viral nephropathy (BK virus, less commonly cytomegalovirus), pyelonephritis, or posttransplant lymphoproliferative disease should be considered. In contrast, histologic evidence of endothelialitis is pathognomonic of rejection and may overlap with antibody-mediated injury.

Acute T-cell–mediated rejection is histologically classified in the Banff criteria by the degree of interstitial inflammation, the quantity of cells infiltrating into tubules, or by endothelialitis of the small arterioles. Type I ACR is characterized by the absence of endothelialitis, with interstitial inflammation of at least 25% of the parenchyma, and tubulitis (Box 109.2 and Fig. 109.5). Type II ACR is characterized by vascular involvement/endothelialitis (Fig. 109.6). Type III ACR is characterized by vascular inflammation that extends to the media (transmural) and may be accompanied by fibrinoid change and necrosis of the smooth muscle cells (Fig. 109.7). Types II and III ACR may or may not be associated with elements of type I ACR; thus the pathologic description of rejection should not be viewed as a pathogenic continuum. However, types II and III ACR often require different therapeutic interventions and carry different prognostic implications than type I ACR (see later sections).

Chronic active T-cell–mediated rejection is a histologic diagnosis that refers to arterial intimal fibrosis specifically with evidence of mononuclear cell infiltration and formation of neointima (Fig. 109.8). This is distinguished from chronic humoral rejection by the location of vascular injury and lack of evidence of pathogenic antibody; it is

BOX 109.1 Banff Classification of Antibody-Mediated Rejection

Active (All Three Criteria Must Be Met for Diagnosis)

- Acute tissue injury with 1 or more of the following:
 - Microvascular inflammation ($g > 0$, $ptc > 0$)
 - Intimal or transmural arteritis ($v > 0$)
 - Acute thrombotic microangiopathy
 - Acute tubular injury
- Evidence of current/recent antibody interaction with vascular endothelium with one or more of the following:
 - Linear peritubular capillary C4d staining
 - Moderate microvascular inflammation ($g + ptc \geq 2$)
 - Increased expression of validated gene transcripts/classifiers in the biopsy tissue strongly associated with antibody-mediated rejection
- Serologic evidence of donor-specific antibodies, including antibodies to human leukocyte antigens and nonhuman leukocyte antigens. C4d staining or expression of validated transcripts/classifiers (criterion 2 earlier) may substitute for donor-specific antibodies.

Chronic Active (All Three Criteria Must Be Met for Diagnosis)

- Morphologic evidence of chronic tissue injury, including 1 or more of the following:
 - Transplant glomerulopathy ($cg > 0$)
 - Severe peritubular capillary basement membrane multilayering on electron microscopy
 - Arterial intimal fibrosis
- Identical to criterion 2 for active antibody-mediated rejection, above
- Identical to criterion 3 for active antibody-mediated rejection, above

Chronic Inactive (All Three Criteria Must Be Met for Diagnosis)

- Transplant glomerulopathy ($cg > 0$) or severe ptc basement membrane multilayering
- Absence of criterion 2 of current/recent antibody interaction with the endothelium
- Prior documented diagnosis of active or chronic active antibody-mediated rejection and/or documented prior donor-specific antibodies

cg, Chronic glomerulopathy; *g*, glomerulitis; *ptc*, peritubular capillaritis; *v*, vasculitis.

Modified from Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney Meeting Report (I): updates on and clarification of criteria for T-cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20:2318–2331.

distinguished from other nonimmunologic processes that may lead to vascular and interstitial fibrosis by the presence of persistent infiltrating cells within vessels. This is covered in greater detail in [Chapter 107](#).

Borderline Rejection

The finding of inflammation in 10% to 25% of the interstitium with tubulitis of less than four mononuclear cells per tubular cross-section is classified as borderline rejection, which remains a pathologic definition without clear clinical significance. When identified in the setting of graft dysfunction or with other findings such as glomerulitis, the risk for progression to clinical rejection on subsequent biopsies is increased and thus treatment may be considered.¹³

CLINICAL MANIFESTATIONS

The clinical presentation of acute rejection is common to both T-cell-mediated and antibody-mediated rejection. Patients typically present

BOX 109.2 Banff Classification of T-Cell-Mediated Rejection

Borderline or Suspicious

Foci of tubulitis ($t1-3$) with mild interstitial inflammation ($i1$) or mild tubulitis ($t1$) with moderate to severe interstitial inflammation ($i1$ or $i2$)

Acute

Grade IA: Interstitial inflammation involving $>25\%$ of nonsclerotic cortical parenchyma ($i > 2$) and moderate tubulitis ($t2$, 4–10 mononuclear cells/tubular cross section)

Grade IB: Interstitial inflammation involving $>25\%$ of nonsclerotic cortical parenchyma ($i > 2$) and severe tubulitis ($t3$, >10 mononuclear cells/tubular cross section)

Grade IIA: Mild to moderate intimal arteritis ($v1$, 0%–25% of luminal area), with or without interstitial inflammation and/or tubulitis

Grade IIB: Severe intimal arteritis ($v2$, $>25\%$ of luminal area), with or without interstitial inflammation and/or tubulitis

Grade III: Transmural arteritis and/or fibrinoid change and necrosis of medial smooth muscle cells with accompanying lymphocyte inflammation ($v3$)

Chronic Active

Grade IA: Interstitial inflammation involving $>25\%$ of sclerotic cortical parenchyma (i -IFTA2 or i -IFTA3) and $>25\%$ of total cortical parenchyma ($ti2$ or $ti3$) with moderate tubulitis ($t2$ or t -IFTA2) involving one or more tubules

Grade IB: Interstitial inflammation involving $>25\%$ of sclerotic cortical parenchyma (i -IFTA2 or i -IFTA3) and $>25\%$ of total cortical parenchyma ($ti2$ or $ti3$) with severe tubulitis ($t3$ or t -IFTA3) involving one or more tubules

Grade II: Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima)

i, Interstitial inflammation; *i*-IFTA, inflammatory interstitial fibrosis and tubular atrophy; *t*, tubulitis; *t*-IFTA, tubulitis in areas of tubular atrophy; *v*, vasculitis.

Modified from Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney Meeting Report (I): updates on and clarification of criteria for T-cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20:2318–2331.

with a rapid rise in serum creatinine and in severe cases may have decreasing urine output, weight gain, fever, or graft tenderness. The clinical findings are commonly nonspecific, and other etiologies of graft dysfunction should be considered at the time of presentation ([Box 109.3](#)) together with the patient's risk for acute rejection ([Box 109.4](#)). As a result of the increase in the number of patients who undergo transplantation despite a risk factor for humoral rejection (e.g., presensitization, known donor-specific antibody in desensitization protocols, and ABO-incompatible transplants), approximately 25% of acute rejection episodes now have a humoral component. In acute humoral rejection there may be features of thrombotic microangiopathy with anemia, evidence of hemolysis, and thrombocytopenia. If there is immediate cyanosis of the graft on revascularization (hyperacute rejection) or an abrupt decline in urine output and graft tenderness 3 to 14 days after transplant (delayed hyperacute or accelerated rejection), donor-specific antibody is implicated. Typically, there is type III ACR and interstitial hemorrhage on biopsy.

PROPHYLAXIS AND PREVENTION

Prophylaxis

The primary goal of transplant management is prevention of immunologic graft loss in the early period after transplant. Over time, the risk for acute rejection diminishes and the approach to immunosuppressive therapy shifts toward considerations of adverse effects of the

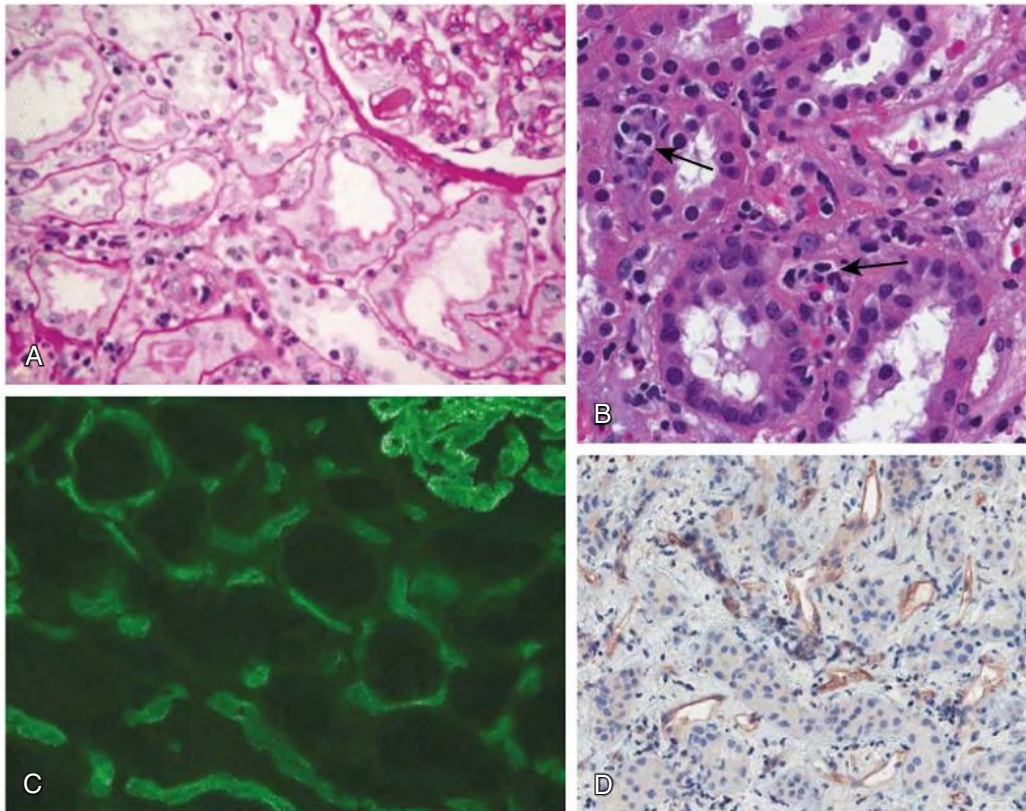


Fig. 109.3 Acute Antibody-Mediated Rejection. (A) Peritubular and glomerular capillaries contain numerous polymorphonuclear leukocytes and mononuclear cells. (B) Numerous polymorphonuclear leukocytes are observed in a peritubular capillary (*arrows*). Interstitial edema is noted. (PAS stain, $\times 200$.) (C) Immunofluorescence staining of peritubular capillaries with C4d. (Fresh frozen tissue sample, $\times 250$.) (D) Immunohistochemistry demonstrating peritubular capillary staining of C4d. (Paraffin-embedded tissue, $\times 480$.) (From Moll S, Pascual M. Humoral rejection of organ allografts. *Am J Transplant*. 2005;5:2611–2618.)

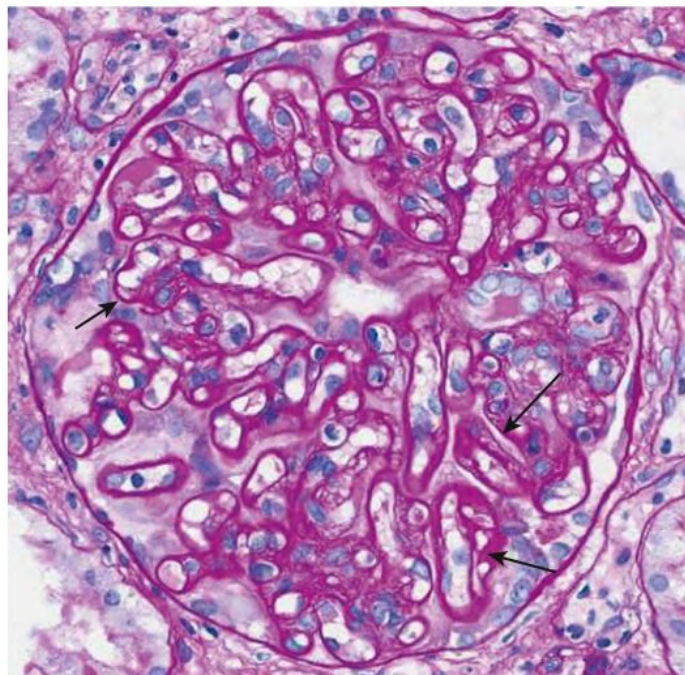


Fig. 109.4 Transplant Glomerulopathy. Light microscopy showing typical membranoproliferative changes, including glomerular basement membrane duplication and thickening (*arrows*).

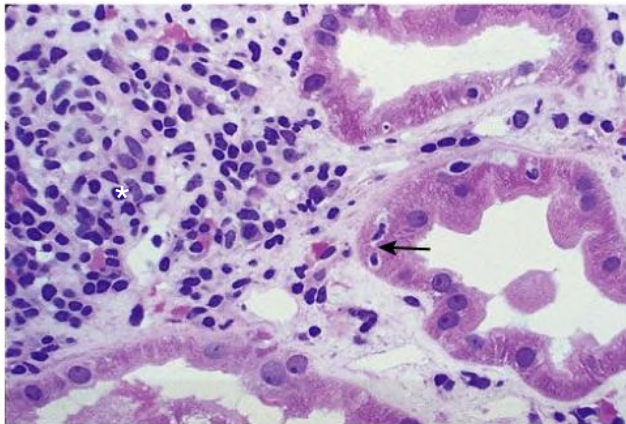


Fig. 109.5 Banff Class I Cellular Rejection. Mononuclear cell infiltrate of the tubules (arrow) and interstitium (asterisk) (Hematoxylin-eosin stain, $\times 200$.) (Courtesy Dr. Maxwell Smith, University of Colorado, Denver.)

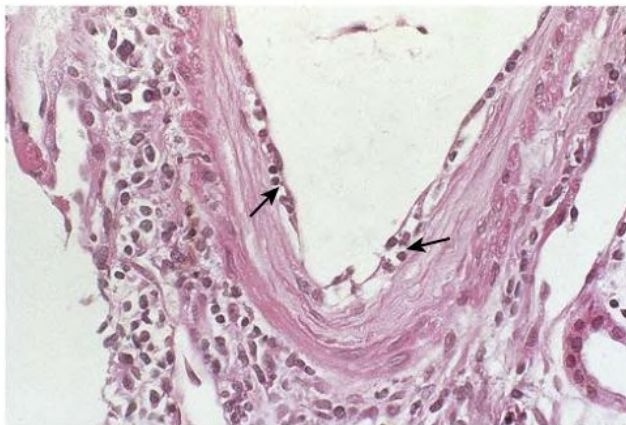


Fig. 109.6 Banff Class II Cellular Rejection. Type II rejection, called acute vascular rejection, manifests with endothelialitis with mononuclear cell infiltration (arrows) beneath the arterial endothelium. (PAS stain, $\times 200$.) (Courtesy Dr. Agnes Fogo, Vanderbilt University, Nashville, TN.)

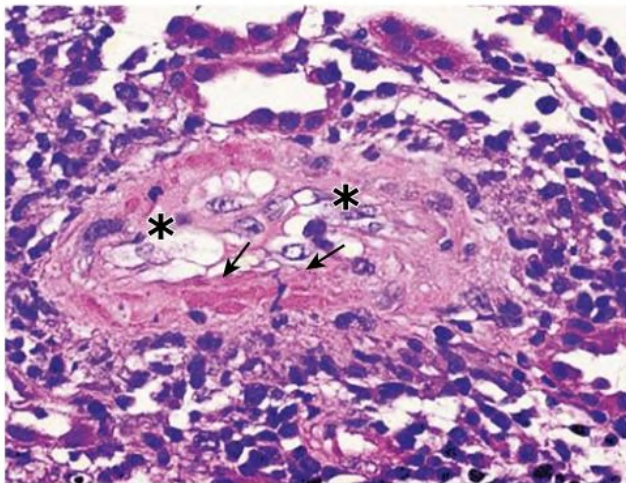


Fig. 109.7 Banff Class III Cellular Rejection. Severe small-vessel vasculitis with transmural mononuclear cell infiltration, fibrinoid necrosis (arrows), and very swollen endothelial cells (asterisks). (PAS stain, $\times 400$.) (From Racusen LC, Colvin RB, Solez K, et al. Antibody-mediated rejection criteria—an addition to the Banff 97 classification of renal allograft rejection. *Am J Transplant.* 2003;3:708–714.)

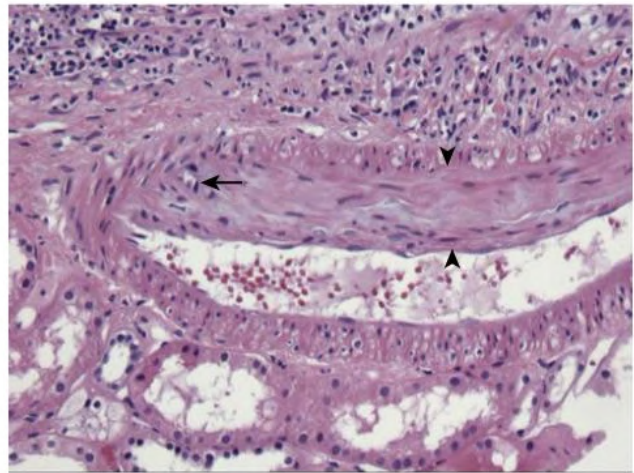


Fig. 109.8 Chronic T-Cell-Mediated Rejection. Chronic allograft arteriopathy with the formation of a fibrous neointima (between arrowheads) and embedded mononuclear cell infiltration (arrow). (Hematoxylin-eosin stain, $\times 200$.) (Courtesy Dr. Maxwell Smith, University of Colorado, Denver.)

BOX 109.3 Differential Diagnosis of Kidney Allograft Dysfunction

Week 1 Posttransplantation

- Acute tubular necrosis
- Hyperacute or accelerated rejection
- Urologic
 - Obstruction
 - Urine leak
- Vascular thrombosis
 - Renal artery
 - Renal vein
- Volume contraction

Less Than 12 Weeks Posttransplantation

- Acute rejection
- Calcineurin inhibitor toxicity
- Volume contraction
- Urologic
 - Obstruction
- Infection
 - Bacterial pyelonephritis
 - Viral infections
- Interstitial nephritis
- Recurrent disease

Longer Than 12 Weeks Posttransplantation

- Acute rejection
- Volume contraction
- Calcineurin inhibitor toxicity
- Urologic
 - Obstruction
- Infection
 - Bacterial pyelonephritis
 - Viral infections
- Chronic allograft nephropathy
- Recurrent disease
- Renal artery stenosis
- Posttransplantation lymphoproliferative disorder

BOX 109.4 Risk Factors for Acute Rejection

High Risk

- Donor-specific HLA sensitization
 - Positive cell-based crossmatch (flow cytometry or CDC)
 - Donor-specific antibody on single-antigen bead analysis
- Delayed graft function
 - Deceased donor source
 - Increased donor age
 - Prolonged ischemic time
 - Donor brain death
 - Donor acute kidney dysfunction
- HLA mismatching
- ABO incompatibility
- Corticosteroid minimization
- Infection
 - Bacterial pyelonephritis
 - Cytomegalovirus
- Adolescent recipient
- Previous rejection episode

Low Risk

- Zero HLA mismatch
- Elderly recipient of young donor kidney
- Preemptive transplantation
- Living donor source
- First transplant

CDC, Complement dependent cytotoxicity; HLA, human leukocyte antigen.

therapy and risks for other events such as cardiovascular (CV) disease and malignancy. Therefore, management includes intensive immunosuppression and monitoring in the first months after transplant with a reduction in the intensity of treatment thereafter.

Prevention of Acute T-Cell–Mediated Rejection: Induction Therapy

A brief course of potent immunosuppression at the time of transplant (referred to as *induction therapy*) is commonly used to prevent acute rejection in transplant recipients, regardless of immunologic risk. According to the Scientific Registry of Transplant Recipients (SRTR), over 90% of US kidney transplant recipients received induction therapy in 2018. The assessment of immunologic risk at the time of transplant has historically considered factors such as overall level of anti-HLA sensitization (panel reactive antibody, or PRA), repeat transplant, African-American race, and recipient age. More recent data suggest pretransplant donor-specific antibodies (DSA) and HLA A/B/DR mismatch to be the main predictors of antibody-mediated rejection and T-cell–mediated rejection, respectively, while PRA and repeat transplantation had no predictive effect¹⁴ (Box 109.4). There continues to be significant variability among transplant centers in the assessment of immunologic risk and its influence on induction immunosuppression prescribing practice. Nevertheless, for higher immunologic risk patients, induction therapy is usually combined with standard doses of immunosuppression to prevent rejection. For those at lower immunologic risk, induction therapy is often employed to minimize exposure to maintenance immunosuppression.

Induction agents can be classified as T-cell depleting or nondepleting. Nondepleting agents include the monoclonal humanized

interleukin-2 (IL-2) receptor antibody (IL-2ra) basiliximab. The IL-2 receptor was identified as a potential immunosuppressive target because it is present on T-cells and inhibition of IL-2/IL-2r signaling inhibits T-cell proliferation (see Chapter 106). Use of IL-2ra has led to lower rates of acute rejection with minimal side effects when combined with cyclosporine-based immunosuppression in the absence of mycophenolate.¹⁵ Importantly, these agents have not been studied in a prospective randomized fashion in combination with a tacrolimus/mycophenolate-based maintenance regimen; thus questions remain regarding the relative benefits of the addition of IL-2ra with more potent, commonly used maintenance agents.¹⁶

Depleting agents include antithymocyte globulin and anti-CD52 monoclonal antibody (alemtuzumab). Unlike the IL-2ra agents, none of these agents have been compared with placebo for the prevention of rejection, and thus their use as induction agents is considered off-label in the United States. Antithymocyte globulin is a polyclonal preparation of antibodies directed at T-cells prepared by immunizing animals with human lymphoid cells derived from the thymus. Currently, the most common antilymphocyte preparation in use is rabbit antithymocyte globulin (rATG). Alemtuzumab binds to CD52, an antigen of unclear physiologic significance that is present on both B and T-cells, and results in depletion of both lymphoid cell lines. Its ability to induce prolonged, significant lymphopenia for up to 6 to 12 months after dosing led to its use in refractory chronic lymphocytic leukemia. Unlike other depleting agents, alemtuzumab is a humanized antibody, and therefore has fewer infusion-related side effects than rATG. Initial trials suggest equivalence to other depleting agents for preventing rejection, but the long-term impact of prolonged lymphopenia on the risk for infection or posttransplant lymphoproliferative disorder has yet to be determined, and comparative trials of induction agents are lacking (see later discussion). Alemtuzumab has been rebranded for use in multiple sclerosis in the United States and is often cost-prohibitive as transplant induction therapy; however, it is still in use in some US and European centers.

Although depleting agents are effective in inhibiting the T-cell response, there is concern about long-term safety. Registry analyses suggest there is an increased risk for future development of lymphoma with depleting agents compared with nondepleting agents or no induction therapy,^{17,18} an association that appears to be dose dependent. Therefore, the potential for recovery or improvement of graft function as a result of repeated or prolonged courses of depleting antibody therapy must be balanced against the potential long-term risk for malignancy.

Three multicenter randomized trials have compared the efficacy of induction agents in the prevention of acute rejection. A multicenter trial in patients at high risk for acute rejection (defined as recipients with an elevated risk for delayed graft function, elevated PRAs, repeat transplants, or HLA mismatches) compared the IL-2ra basiliximab to rATG with maintenance immunosuppression of cyclosporine, mycophenolate mofetil (MMF), and prednisone.¹⁹ At 12 months, the rate of acute rejection in the basiliximab cohort was 26% versus 16% in the rATG arm. A second trial in patients at high risk for acute rejection (defined by elevated PRAs, repeat transplants, or loss of a previous kidney transplant to acute rejection) compared the IL-2ra daclizumab to rATG, with a remarkable increase in the risk for rejection for patients randomized to daclizumab (27%) compared with rATG (15%) at 1 year.²⁰ Thus, for patients at increased risk for rejection, rATG provided greater prevention of acute rejection, supporting its use in higher risk populations. In contrast, the European multicenter randomized controlled HARMONY trial showed no difference in the proportion of low immunologic risk patients with acute rejection receiving either rATG followed by rapid steroid withdrawal or basiliximab with or without

rapid steroid withdrawal.²¹ Finally, a clinical trial compared anti-CD52 to either rATG in high-risk patients (repeat transplant, elevated PRAs, or Black ethnicity) or to an IL-2ra in low-risk patients, followed by early corticosteroid withdrawal and maintenance tacrolimus and mycophenolate immunosuppression.²² Acute rejection rates over the first 12 months in the high-risk arms were 10% for anti-CD52 and 13% for rATG and in the low-risk arms were 20% for IL-2ra and 3% for anti-CD52. However, acute rejection rates in anti-CD52-treated patients were higher in both low- and high-risk arms, suggesting a delayed (>12 months) immunologic risk for immune reconstitution with anti-CD52 therapy. This may be problematic for transplant care providers who generally adhere to less rigorous late (>12 months) clinical monitoring protocols for otherwise stable transplant recipients. Therefore, we recommend adhering to a more frequent late (>1 year) clinical monitoring protocol for patients receiving induction with alemtuzumab.

In summary, it remains unclear whether induction therapy is required in low-risk patients pending further prospective head-to-head trials, whereas patients at higher immunologic risk appear to benefit from lymphocyte-depleting treatment. Based on SRTT data in 2018, approximately 75% of US transplant recipients received rATG induction and 20% received IL-2ra induction, with the remaining cohort receiving either no induction or other agents. Induction regimens are commonly protocolized at individual transplant centers by balancing immunologic protection against cost and the risk of adverse effects. Common dosing regimens of induction agents are listed in Table 109.1.

Prevention of Acute Antibody-Mediated Rejection: Desensitization and Paired Exchange

Nearly all patients with high levels of donor-specific antibodies before transplant or who are blood type-incompatible to the donor before

transplant will develop acute antibody-mediated rejection after transplant without pretransplant intervention such as desensitization protocols. Desensitization protocols typically involve removal of preformed antibody with plasma exchange and/or suppression of antibody production and action with intravenous immunoglobulin (IVIG) or B-cell inhibition with rituximab. Choice of desensitization strategy is influenced by factors such as type of transplant (living vs. deceased donor) and the degree of HLA-antibody reactivity one is attempting to overcome. In the case of living donation the crossmatch barriers are known in advance, so plasma exchange followed by low-dose (100–200 mg/kg) IVIG can be arranged followed by transplantation when the desired crossmatch results are obtained. Protocols consisting of monthly high-dose IVIG (1–2 g/kg) infusions with or without rituximab²³ have been used when pretransplant plasma exchange is not feasible, such as in sensitized patients without living donor options awaiting a deceased donor kidney, but the effectiveness of this strategy has been questioned.²⁴ Published desensitization agents and protocols are listed in Table 109.2.

Desensitization strategies allow transplantation to proceed in many sensitized patients, but are associated with high rates of acute rejection, ranging from 20% to 70% (depending factors including specific protocol, induction immunosuppression, and immunologic risk), which are often humoral and frequently progress to chronic antibody-mediated injury.^{25,26}

Kidney paired exchange programs increasingly have been used over the past 10 years in Europe and the United States and allow patients with existing HLA- or ABO-incompatible living donors to receive a compatible transplant while avoiding toxicities and long-term immunologic damage associated with desensitization protocols. Donor-recipient “chains” are usually initiated by a single nondirected live donor and can result in multiple patients with incompatible live donors receiving HLA- and ABO-compatible organs. According to the US

TABLE 109.1 Agents Used for Induction and Treatment of T-Cell-Mediated Rejection

Agent	Target	Dose (Induction)	Dose (Rejection)
Methylprednisolone	B cells, T-cells, macrophages	250–500 mg IV with induction therapy, followed by taper over 1–5 days	3–5 mg/kg (250–500 mg) IV × 3–5 days
Basiliximab	IL-2 receptors on cells	20 mg IV × 2 on days 0 and 4	N/A
rATG	T-cell surface antigens	1–1.5 mg/kg IV × 4–14 days	1–1.5 mg/kg IV × 7–14 days
Alemtuzumab	CD52 on T and B cells	30–60 mg × 1 or 2 on days 0 and 2	Same as induction

IL-2, Interleukin-2; IV, intravenous; N/A, not applicable; rATG, rabbit antithymocyte globulin.

TABLE 109.2 Agents Used for Desensitization and Treatment of Antibody-Mediated Rejection

Treatment	Mechanism	Protocol (Desensitization)	Dose (Antibody-Mediated Rejection)
Plasma exchange ^a	Antibody removal	2–4 sessions or until XM acceptable, combined with IVIG	2–5 treatments, daily or every other day, combined with IVIG
IVIG ^a	Multiple, antibody inhibition(?)	100–200 mg/kg after plasma exchange until acceptable XM or 1–2 g/kg/mo until transplant	100–200 mg/kg after plasma exchange
Rituximab ^a	Anti-CD20 B-cell inhibition	375 mg/m ² (day 15) combined with IVIG 1–2 g/kg (days 1 and 30)	375 mg/m ² with plasma exchange and IVIG
Bortezomib	Plasma cell inhibition	Not established	1.3 mg/m ² × four doses over 1–2 wk, usually combined with plasma exchange and IVIG
Eculizumab	Terminal complement C5 inhibition	Not established	For prevention in + XM transplant: 600–1200 mg/wk × 4 then biweekly until successful antibody reduction
Splenectomy	B-cell removal	No longer used	N/A, for severe refractory cases only

^aCommonly used.

IVIG, Intravenous immunoglobulin; N/A, not applicable; XM, crossmatch.

Renal Data System, paired exchange accounted for 14% of living donor transplants in 2018, up from 5% in 2009. Recent data show no difference in graft or patient survival at 3, 5, and 7 years between patients receiving paired exchange transplants through the US National Kidney Registry versus those receiving direct living donor transplants in the United States, despite a higher burden of risk factors in the patients transplanted through paired exchange.²⁷

Maintenance Therapy for Prevention of Acute Rejection

Current maintenance immunosuppression most commonly includes a CNI, an antiproliferative agent, and corticosteroids. This combination of agents forms the standard against which novel strategies are compared, such as corticosteroid withdrawal/avoidance and CNI withdrawal/avoidance. Cyclosporine has been replaced by tacrolimus as the preferred CNI in the United States and, together with the antiproliferative agent MMF, it forms the most common immunosuppressive regimen in current practice in most high-income countries (Table 109.3). More affordable regimens that include cyclosporin A (CsA) and azathioprine (AZA) are commonly used in areas where resources are more limited, but appear to lead to higher rates of rejection.²⁸

Calcineurin Inhibitors in the Prevention of Acute Rejection

Since the early 1980s, when the introduction of CsA led to reduced risk of acute rejection and to better graft survival,²⁹ calcineurin inhibition has been a cornerstone of maintenance immunosuppression. Tacrolimus, first introduced in 1990s and compared head-to-head with CsA in multiple trials, provides greater protection from acute rejection but with a different side effect profile. A recent meta-analysis of trials demonstrated that compared with CsA, tacrolimus reduced the risk of acute rejection by 31% but increased the risk of new diabetes by 86%. Tacrolimus was also associated with better death-censored graft survival (hazard ratio, 0.56), particularly at target trough doses less than 10 ng/mL, a finding that was not shown in individual studies.³⁰

Antiproliferative Agents in the Prevention of Acute Rejection

AZA was the first antiproliferative agent used in kidney transplantation and was used initially in conjunction with corticosteroids and later with CsA. Although its development was critical in the advancement of allotransplantation, acute rejection was quite common, with acute rejection rates of 35% to 40% in a number of clinical trials using CsA/AZA/prednisone. Newer antiproliferative agents emerged in the 1990s with MMF and later with the mammalian target of rapamycin (mTOR) inhibitors sirolimus and everolimus, which significantly reduced the incidence of acute rejection.

MMF is a purine antagonist that interferes with DNA synthesis in rapidly dividing cells such as activated lymphocytes, and it gained popularity after a number of multicenter prospective clinical trials demonstrated that it reduced the risk of acute rejection by approximately 50% compared with AZA.^{31,32} A recent meta-analysis of 23 studies showed MMF to be associated with significantly fewer cases of acute rejection and graft loss compared with AZA.²⁸ Most studies in this analysis reported outcomes of MMF and AZA in combination with CsA and thus may not translate to regimens using tacrolimus. One drawback of MMF has been its poor gastrointestinal (GI) tolerability, which often results in reduction of therapy with attendant risks for acute rejection³³ and graft loss.³⁴ An MMF analog, enteric-coated mycophenolate acid (MPA), has been developed and appears to be “noninferior” in efficacy to MMF,³⁵ with fewer GI side effects reported in one open-label study.³⁶

Similar to MMF, the mTOR inhibitors sirolimus and everolimus were initially tested in clinical trials as a substitute for AZA and reduced the risk of acute rejection to a similar extent as MMF.³⁷ mTOR inhibitors inhibit the progression from G1 to S phase of the cell cycle and appear to have additional antiproliferative effects on nonimmune cells that may increase the risk of side effects (impaired wound healing, lymphocyte formation, proteinuria, and slower recovery from delayed graft function) but may reduce the incidence of viral infection and malignancy.^{38,39} mTOR inhibitors also augment the nephrotoxic effects of CNI, and therefore a reduced dose of CNI is advisable when used together with mTOR inhibitors. A recent large trial (TRANSFORM) randomized 2037 subjects to reduced-dose CNI, most commonly tacrolimus (tacrolimus C₀ 2–4 ng/mL) in combination with everolimus (everolimus C₀ 3–8 ng/mL), or standard CNI (e.g., tacrolimus C₀ 6–10 ng/mL) with mycophenolate.⁴⁰ At 12 months posttransplant, noninferiority was demonstrated between treatment arms for the combined endpoint of treated biopsy-proven acute rejection or estimated glomerular filtration rate (eGFR) less than 50 mL/min/1.73 m² (48.2% in everolimus arm vs. 45.1% in mycophenolate arm) with no difference in graft loss or death but less cytomegalovirus and BK virus infections in the everolimus arm.

Acute Rejection Rates in Calcineurin-Sparing and Corticosteroid-Sparing Immunosuppression Regimens

There has been increasing interest in how best to minimize side effects of immunosuppression. Avoidance of CNIs offers the hope of prolonged graft survival given the inherent nephrotoxicity of these medications, and avoidance of corticosteroids offers the hope of reducing a number of cosmetic, metabolic, and CV side effects attributable to prednisone.

In the United States approximately one-third of patients undergo early corticosteroid cessation (within 7 days after transplant).⁴¹ Generally, patients at lower immunologic risk (low PRAs, first transplants) are selected,^{42,43} and immunosuppression includes induction therapy, a CNI, and an antiproliferative agent. The proportion of patients with acute rejection in single-center studies ranges from 10% to 15%. In the largest prospective, double-blind multicenter study to date of corticosteroid cessation versus standard corticosteroid maintenance,⁴⁴ a standard steroid taper or a rapid elimination of steroids at 7 days posttransplant was compared on the background of induction therapy plus a tacrolimus/MMF-based immunosuppression. Patient survival, graft survival, and creatinine clearance were comparable at 5 years. CV risk factors and weight gain were not significantly different between groups, but corticosteroid withdrawal was associated with less bone disease, less insulin-requiring new-onset diabetes, and lower triglyceride levels. One cause for concern with corticosteroid withdrawal was that rejection rates were higher (18% vs. 11%,

TABLE 109.3 Immunosuppression Regimens in the United States, 2018

Regimen	Percent (Approximated)
Induction Regimens at Time of Transplant	
IL-2 receptor antibody	20
T-cell depleting	72
Maintenance Regimens at Time of Hospital Discharge	
TAC/MMF-MPA/Pred	65
TAC/MMF-MPA	30
Other	5

MMF, Mycophenolate mofetil; MPA, mycophenolate acid; Pred, prednisone; TAC, tacrolimus.

Modified from OPTN/SRTR Annual Report, 2020.

$P = .04$), and a post hoc analysis suggested a higher proportion of chronic allograft nephropathy in the corticosteroid withdrawal arm. In contrast, the aforementioned HARMONY study reported similar rejection rates and significant reductions in posttransplant diabetes in two rapid (8-day) steroid withdrawal arms (one using basiliximab and the other using rATG induction) compared with the control arm of basiliximab induction and maintenance steroids, with all patients receiving tacrolimus and MMF.²¹

Clinicians must weigh the possibility of higher rejection risk versus the potential benefits when counseling patients regarding corticosteroid withdrawal. Consideration of corticosteroid withdrawal is best supported for the patient with a lower risk of rejection (low immunologic risk, expected immediate graft function) or increased risk of corticosteroid-related complications such as bone disease and diabetes.

CNI avoidance has been studied with both dual therapy (MMF/prednisone) and triple therapy with two antiproliferative agents (sirolimus/MMF/prednisone) in combination with induction therapy. In general, MMF/prednisone maintenance immunosuppression does not appear to effectively prevent rejection (70% incidence in one pilot study⁴⁵), and although single-center studies report acute rejection rates of 6% to 13% with sirolimus/MMF/prednisone therapy,^{46,47} a large multicenter trial using this combination and target trough concentrations of sirolimus 4 to 8 ng/mL also revealed an excessively high acute rejection rate (38%).⁴⁸ CNI avoidance with the costimulation blocker belatacept also is associated with a higher proportion of early acute rejection than CNI-based regimens but still may better protect glomerular filtration rate (GFR) over the longer term. In the BENEFIT trial, which compared 2 dosing regimens of belatacept (low and moderate intensity) to CsA with IL-2ra induction and MPA/prednisone maintenance,⁴⁹ higher 1-year acute rejection rates were noted in the belatacept arms (22% and 17% for higher and lower intensity, respectively, vs. 7% for CsA) but graft survival was comparable and measured GFR was significantly higher for those receiving belatacept (65 and 63 mL/min for high- and low-intensity belatacept vs. 50 mL/min for CsA). With follow-up to 7 years, the belatacept arms demonstrated a 43% reduced risk for a composite endpoint of death or graft loss compared with CsA.⁵⁰ Importantly, patients with low immunologic risk were selected for study, and excluded those who were Epstein-Barr virus (EBV)-seronegative given higher rates of posttransplant lymphoproliferative disorder associated with belatacept use in previous reports.⁵¹ A post hoc analysis of patients included in both the BENEFIT and BENEFIT-EXT studies (the latter including patients receiving extended-criteria donor kidneys) showed significantly lower blood pressure, serum lipids, and incidence of new-onset diabetes in those randomized to belatacept regimens compared with CsA.⁵² We suggest that EBV-seronegative patients at low immunologic risk, in particular those with pretransplant metabolic profiles at risk for posttransplant metabolic complications, may benefit from belatacept-based CNI-free immunosuppression.

In contrast to their use in strategies, that avoid CNI, mTOR inhibitors have achieved relative success when used in CNI withdrawal^{53,54} and conversion protocols. In one study, patients receiving a CNI, MMF, and prednisone were randomized to continue this regimen or undergo CNI conversion to sirolimus at 1 to 6 months after transplant with statistically similar rates of acute rejection and comparable kidney function at 2 years.⁵⁵ Another study, employing a similar approach of CNI conversion to everolimus at 4 to 5 months posttransplant, demonstrated equivalent rates of acute rejection (15% in both groups) and significantly better eGFR at 12 months in the conversion group (72 mL/min vs. 62 mL/min in the CNI continuation arm).⁵⁶ A recent meta-analysis that included 23 studies comparing CNI conversion to mTOR inhibitors found a modest improvement in GFR with no

difference in rates of rejection or graft loss.⁵⁷ Widespread acceptance of CNI to mTOR inhibitor conversion strategies has been hindered because of mTOR inhibitor-related side effects and reluctance to alter immunosuppression in stable transplant recipients.

Ideally, a strategy of combined CNI and steroid avoidance would eliminate the chronic accumulation of side effects and toxicities caused by these agents. A recent trial using depleting antibody induction (alemtuzumab or rATG) and belatacept/MMF-based immunosuppression combined with early corticosteroid withdrawal demonstrated comparable kidney function and patient and graft survival compared with rATG, tacrolimus/MMF-based immunosuppression with early corticosteroid withdrawal.⁵⁸ However, 1-year acute rejection rates were still higher in the belatacept cohorts (15.9% with alemtuzumab, 22.1% with rATG) compared with the tacrolimus cohort (4.8%, $P = .024$ and $< .001$, respectively). Thus, there is still room for improvement and additional immunosuppression strategies are needed to minimize side effects while preventing alloimmunity. Acute rejection rates by treatment regimen reported in recent multicenter clinical trials are shown in Table 109.4.

TREATMENT

Acute T-Cell–Mediated Rejection

Treatment of T-cell–mediated acute rejection is often directed by the findings on biopsy and the clinical response to pulse corticosteroids. For the patient with graft dysfunction and biopsy-proven rejection, treatment with intravenous methylprednisolone 3 to 5 mg/kg (250–500 mg/day) for 3 to 5 days is often effective if the histologic injury is tubulointerstitial (Banff class IA or IB). Remarkably few studies of the clinical response to corticosteroids in the treatment of acute rejection have been performed under modern immunosuppression, but prior data suggest that 60% to 70% of patients will respond with improved urine output and decreasing serum creatinine within 5 days. If there is inadequate response after corticosteroid pulse therapy or if there is vascular involvement (Banff class IIA, IIB), corticosteroids often must be supplemented with T-cell–depleting antibody therapies in a similar dosing strategy but longer treatment course compared with their use for induction (see Table 109.1). The duration of treatment has typically been 7 to 14 days, with no clinical trials investigating the efficacy of shorter courses versus longer courses. For patients who are on a maintenance regimen that is not tacrolimus based, tacrolimus conversion may be considered in the setting of rejection with an inadequate response to corticosteroids,⁵⁹ whereas for patients on a corticosteroid-free regimen, reinstitution of maintenance prednisone may be warranted.⁶⁰ For patients who do not respond adequately to corticosteroids, T-cell–depleting agents are required. A typical course of rATG for T-cell–mediated rejection is 1.5 mg/kg for 4 to 14 days dictated by clinical response and absolute lymphocyte counts, again with no controlled study comparing one treatment regimen versus another.

Acute Antibody-Mediated Rejection

Treatment of acute humoral rejection is indicated when the triad of graft injury, evidence of antibody/endothelium interaction, and circulating donor-specific antibody is present, but should also be considered in high-risk circumstances (prior desensitization or known donor-specific antibody) even if all three criteria are not met. High-quality randomized trials investigating treatment options for acute humoral rejection are lacking,⁶¹ and strategies are generally dictated by center experience. Traditionally, treatment has entailed plasma exchange (to remove the pathogenic immunoglobulin[s]) and IVIG (to inhibit/suppress antibody production). In general, at least 5 plasma exchange treatments should be administered with 1 to 2 g/kg total dose

TABLE 109.4 Maintenance Immunosuppression and Reported Rejection Rates in Randomized Multicenter Trials

Regimen	Induction	CNI Dose or Trough Level Goal (ng/mL)	Antiproliferative Dose or Trough Level Goal	Acute Rejection at 6 Months (%)
Low Immunologic Risk				
CsA/AZA/Pred		4 mg/kg/d	1.5–2 mg/kg/day	36
CsA/AZA/Pred	IL-2ra	Not stated	Not stated	22
CsA/MMF/Pred		150–300 × 3 mo, 100–200	1 g bid	24
CsA/MMF/Pred	IL-2ra	125–400 × 3 mo, 100–300	1 g bid	12
TAC/AZA/Pred	OKT3 or Atgam	5–14	1.5 mg/kg/day	32
TAC/MMF/Pred	OKT3 or Atgam	5–14	1 g bid	7
TAC/MMF/Pred	IL-2ra	7–16 × 3 mo, 5–15	1 g bid	4
CsA/SRL/Pred		200–350 × 1 mo, 200–300 × 1 mo, 150–250	2 mg/day	17
TAC/SRL/Pred		8–16 × 3 mo, 5–15	4–12 ng/mL	13
High Immunologic Risk				
CsA/SRL/Pred	IL-2ra or rATG	200–300 0–14 days 150–200	10–15 ng/mL	14 ^a
TAC/SRL/Pred	IL-2ra or rATG	10–15 0–14 days 5–10	10–15 ng/mL	17
Drug Minimization or Avoidance				
CsA (low)/MMF/Pred	IL-2ra	50–100	1 g bid	23
TAC (low)/MMF/Pred	IL-2ra	3–7	1 g bid	12³
TAC/MMF	IL-2ra or rATG	10–20 × 3 mo, 5–15	1.5 g bid × 14 days 1 g bid	9⁴
SRL/MMF/Pred	IL-2ra	N/A	SRL 4–8 ng/mL MMF 1 g bid	38
Bela/MMF/Pred	IL-2ra	N/A	1 g bid	22 (MI) 17 (LI)^{49,50}
CNI → SRL/MMF/Pred	IL-2ra or rATG or OKT3	N/A	SRL 5–10 ng/mL MMF 1–1.5 g bid	7.4 ^a
CsA → EVL/MMF/Pred	IL-2ra	N/A	EVL 6–10 ng/mL MMF 1 g bid	15 ⁵
TAC (low)/EVL/Pred	IL-2ra or rATG	Month 0–2; 3–6; >6: TAC 4–7; 2–5; 2–4	EVL 3–8 ng/mL	10⁶
Bela/MMF	Anti-CD52 or rATG	N/A	MMF 1 g bid	16 ⁶ (anti-CD52); 22 ^a (rATG)

Bold denotes suggested regimens by the authors based on available data.

^aAcute rejection rate at 12 months.

AZA, azathioprine; Bela, belatacept; bid, twice daily; CNI, calcineurin inhibitor; CsA, cyclosporine A; EVL, everolimus; IL-2ra, interleukin-2 receptor antibody; LI, less intensive; MI, more intensive; MMF, mycophenolate mofetil; N/A, not applicable; Pred, prednisone; rATG, rabbit antithymocyte globulin; SRL, sirolimus; TAC, tacrolimus.

IVIG. IVIG is removed by plasma exchange, so a common strategy is to administer IVIG 100 to 200 mg/kg after each exchange. For refractory acute humoral rejection, rituximab may be considered despite targeting B-cells at an earlier phase of maturation than the antibody-producing plasma cell line.⁶² The largest randomized clinical trial of rituximab treatment for antibody-mediated rejection to date found no difference between the treatment group and placebo when added to plasma exchange and IVIG for a composite endpoint of early improvement in kidney function at 12 days after treatment and graft loss at 1 year. However, the study was limited by a small sample size ($n = 38$) and significant crossover between groups with 42% of subjects in the placebo arm receiving rescue doses of rituximab.⁶³ In contrast, the proteasome inhibitor bortezomib directly inhibits antibody-producing plasma cells and has been reported in single-center case series as a potential treatment for refractory antibody-mediated rejection, usually in combination with other modalities such as plasma exchange and IVIG.⁶⁴ Eculizumab is a humanized monoclonal antibody that blocks terminal complement activation and has been shown to significantly reduce the risk for acute humoral rejection, as well as subsequent transplant glomerulopathy in patients who have undergone positive

crossmatch kidney transplant⁶⁵; however, feasibility of this approach is limited by high cost. Targeting the initiation of the classical complement cascade with C1 inhibition has also gained recent interest; pilot studies suggest that C1 inhibition may be associated with improved eGFR and a reduced risk of development of transplant glomerulopathy.^{66,67} Finally, there are case reports of splenectomy for refractory acute humoral rejection.⁶⁸ These therapies are typically coupled with targeted T-cell therapy such as high-dose steroids and/or depleting antibody therapy, because histologic evidence of mixed T-cell rejection is often present and helper T-cell function may contribute to an enhanced B-cell response. Agents used for treatment of acute humoral rejection are summarized in Table 109.2.

Chronic Rejection (T-Cell–Mediated and/or Antibody Mediated)

T-cell–mediated or antibody-mediated injury in a graft without features of acute tissue injury remains a therapeutic dilemma. No intervention has been proven effective in reversing the chronic tissue injury; transitioning to tacrolimus/MMF therapy or increasing the dose of these agents should be considered if CNI nephrotoxicity is not

identified.⁶⁹ Any intervention should be weighed against the potential for risk for enhanced immunosuppression and the lack of long-term data. This topic is discussed in greater detail in [Chapter 112](#).

PROGNOSIS

Acute rejection may predispose to chronic graft dysfunction, with increased histologic findings of chronic rejection and/or interstitial fibrosis and tubular atrophy and clinical findings of reduced graft survival. The clinical response to antirejection therapy appears to be critical in this regard because the change in kidney function from 6 and 12 months posttransplant is more predictive of long-term graft survival than the occurrence of prior episodes of acute rejection.⁷⁰ Two analyses (one examining the US experience⁷¹ and another examining the Australia/New Zealand experience⁷²) have shed light on risk factors for graft loss after episodes of acute rejection. In general, acute T-cell-mediated rejection that responds to therapy with return to near baseline kidney function does not portend worse graft survival. However, vascular rejection, late rejection (after 3 months), and rejection that does not respond to within 75% of baseline serum creatinine is associated with worse graft outcomes. Although the risk for acute rejection has fallen significantly over the past decade, the likelihood of graft survival has not improved in parallel, possibly because contemporary rejection is less responsive to therapy, with fewer cases achieving near-baseline serum creatinine levels. The long-term prognosis after episodes of acute antibody-mediated rejection has not been fully defined in prospective analyses. However, it appears that episodes of acute humoral rejection likely affect long-term graft survival. For example, a retrospective analysis of 302 patients with biopsy-proven acute rejection (192 with T-cell-mediated rejection and 110 with antibody-mediated rejection) found a threefold to ninefold increased risk for graft loss at 5 years associated with antibody-mediated rejection compared with T-cell-mediated

rejection.¹⁰ Similarly, the emergence of de novo HLA antibodies at any time after transplant is associated with a 5% worse graft survival per year compared with those who do not form anti-HLA antibodies.⁷³ For this reason, patients who have had episodes of acute rejection must be rigorously monitored with optimization of baseline maintenance immunosuppression. Remaining questions include the value of escalated immunosuppression, such as longer-term scheduled antibody therapy and additional IVIG treatment for those without adequate clinical response or with persistently elevated titers of HLA antibodies.

SUMMARY AND RECOMMENDATIONS

The incidence of acute rejection has decreased over time, but it remains an important cause of graft dysfunction and progressive graft loss, particularly when antibody-mediated forms of injury are identified. Higher-risk patients benefit from induction therapy with lymphocyte-depleting agents for the prevention of rejection. Although many maintenance immunosuppressive regimens have been used to minimize the incidence of rejection and side effects, the most effective maintenance immunosuppressive regimen for the prevention of rejection is a three-drug regimen consisting of tacrolimus, mycophenolate, and prednisone. Alternative immunosuppressive strategies may be considered for side effects or toxicities related to immunosuppressive agents, which in clinical practice has led to a myriad of treatment combinations (see [Table 109.4](#)). Steroid minimization and belatacept-based CNI-free immunosuppression have been associated with adequate short- and intermediate-term outcomes but should be limited to those patients at lower immunologic risk for acute rejection. Until it is possible to determine an individual patient's degree of immune function, suppression, or graft-specific immunity, the clinician must determine the best treatment regimen based on potency (potential to minimize rejection) and tolerability.

SELF-ASSESSMENT QUESTIONS

- According to the Banff Working Classification of Renal Allograft Pathology, which one of the following patterns would be recognized as acute antibody-mediated rejection?
 - Circulating donor-specific antibodies, severe tubulitis, interstitial inflammation
 - Peritubular capillaritis (ptc1), interstitial inflammation, moderate transplant glomerulopathy (cg2)
 - Circulating donor-specific antibodies, mild to moderate large-vessel intimal arteritis, moderate interstitial fibrosis
 - Peritubular capillary C4d deposition, circulating donor-specific antibodies, peritubular capillaritis (ptc2)
- Which one of the following statements is *true* regarding induction immunosuppression for the prevention of acute rejection?
 - rATG is a monoclonal antibody directed against the CD3 subunit of the T-cell receptor associated with a significant innate immune response resulting in a cytokine release syndrome.
 - Lymphocyte-depleting agents are more effective in preventing acute rejection compared with IL-2 receptor antibodies in patients at high immunologic risk.
 - The risk for posttransplant lymphoproliferative disorders is higher in patients treated with IL-2 receptor antibodies than in those treated with lymphocyte-depleting agents.
 - Patients at low immunologic risk experience higher early (<12 months) rates of acute rejection when treated with anti-CD52 therapy (Campath) compared with IL-2 receptor antibody therapy.
- Which one of the following is the *most* effective maintenance immunosuppressive regimen for the prevention of rejection?
 - Sirolimus, MMF, prednisone
 - Tacrolimus, MMF, prednisone
 - Belatacept, MMF, prednisone
 - Tacrolimus, MMF, prednisone withdrawal at 7 days posttransplant
- A 47-year-old man treated with tacrolimus, MMF, and 5 mg of prednisone after undergoing deceased-donor kidney transplant 5 months prior presents with an elevated serum creatinine of 5.2 mg/dL (recent baseline 1.3 mg/dL) and mild graft tenderness. Kidney ultrasound is nondiagnostic, a workup for infectious causes is negative, and an assay for donor-specific antibody is negative. Allograft biopsy is performed and reveals severe tubulitis with vascular intimal arteritis without evidence of peritubular capillary C4d deposition. Which of the following is the *most* appropriate initial therapy?
 - Increasing the dose of maintenance prednisone to 60 mg/day for 7 days followed by a dose taper back to 5 mg
 - 3 to 5 days of intravenous methylprednisolone at 500 mg/day
 - rATG 1.5 mg/kg IV for 7 to 14 days
 - Plasmapheresis followed by 100 to 200 mg/kg immune immunoglobulin for a total of 5 sessions

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